

Some counterexamples on higher order Bergman kernels

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Abstract

We present five counterexamples in the theory of higher order Bergman kernels. The first answers in the negative a convexity question of Błocki–Zwonek; the second concerns the equality case of the higher order Suita conjecture; and the third shows that a natural identity between higher order p -Bergman kernels and ξ - p -Bergman kernels fails for $0 < p < 1$. The fourth shows that the monotonicity of the higher order p -Bergman kernels along the Green sublevel sets, which holds for $p \in [1, 2]$ by the results of Błocki–Zwonek and Bao–Guan–Sun, fails for every $p > 2$. The fifth shows that the L^p analogue of Guan’s concavity theorem for minimal L^2 integrals on sublevel sets of plurisubharmonic functions fails for every $p > 2$. The constructions and proofs of these counterexamples come from AI tools.

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The first counterexample presented in this paper was originally produced by GPT 5.6 sol xhigh, the second and the third counterexamples were produced by Codex with Deepseek v4 flash, and the fourth and the fifth counterexamples were produced by Codex with Deepseek v4 pro. The author has independently verified the statements and all computations.

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1 Introduction

Let $D \subset \mathbb{C}^n$ be a bounded pseudoconvex domain and let $w \in D$. The Bergman kernel of D at w is defined by

$$K_D(w) := \sup \left\{ |f(w)|^2 : f \in \mathcal{O}(D), \|f\|_D := \left(\int_D |f|^2 d\lambda \right)^{1/2} \leq 1 \right\},$$

where $d\lambda$ denotes the Lebesgue measure. Following [4], for a homogeneous polynomial H of degree k , $H(z) = \sum_{|\alpha|=k} a_\alpha z^\alpha$, we write $P_H(f) := \sum_{|\alpha|=k} a_\alpha D^\alpha f$, and we define the higher order Bergman kernel

$$K_D^H(z) := \sup \left\{ |P_H(f)(z)|^2 : f \in \mathcal{O}(D), f^{(j)}(z) = 0 \ (0 \leq j < k), \|f\|_D \leq 1 \right\}.$$

In dimension one, $H(z) = z^k$ and $K_D^{z^k}(z)$ will also be denoted by $K_D^{(k)}(z)$.

Let $G_D(\cdot, w)$ be the (pluricomplex) Green function of D with pole at w , and assume $w = 0$. For $a \leq 0$ we denote

$$D_a := e^{-a} \{ G_D(\cdot, 0) < a \}.$$

The main theorem of [4] asserts that $a \mapsto K_{D_a}^H(0)$ is non-decreasing, and it is noted that

$$K_{D_a}^H(0) = e^{2(n+k)a} K_{\{G_D(\cdot, 0) < a\}}^H(0).$$

The five counterexamples in this paper concern five natural extremal questions for these kernels.

1.1 The convexity question of Błocki–Zwonek

The main theorem of [4] also motivates the following question raised in [4, Remark 3].

QUESTION 1.1 (Błocki–Zwonek, [4, Remark 3]). *Is the function*

$$(-\infty, 0] \ni a \longmapsto \log K_{D_a}^H(w)$$

convex, as it is in the case $H \equiv 1$?

The convexity in the case $H \equiv 1$ follows from Berndtsson's theorem on the plurisubharmonic variation of Bergman kernels (see the final remark in [2]). Our first result shows that Question 1.1 has a negative answer in general.

THEOREM 1.2. *For $Q > 1$ sufficiently large set $c := 2Q$ and*

$$D_c := \{z \in \mathbb{C} : |z|^2 < |1 - cz^2|\}.$$

Then D_c is a domain in $\mathbb{C}$ (indeed $\mathbb{C}$ minus two disjoint Jordan regions, hence unbounded and doubly connected), $0 \in D_c$, and with $H(z) = z^6$ the function

$$(-\infty, 0] \ni a \mapsto \log K_{(D_c)_a}^{z^6}(0)$$

is not convex. More precisely, for

$$a_- := -\log 2, \quad a_0 := -\frac{1}{2} \log 2, \quad a_+ := 0,$$

so that $a_0 = (a_- + a_+)/2$, we have

$$\log K_{(D_c)_{a_0}}^{z^6}(0) > \frac{1}{2} \left(\log K_{(D_c)_{a_-}}^{z^6}(0) + \log K_{(D_c)_{a_+}}^{z^6}(0) \right),$$

and the difference is

$$\frac{3}{64Q^2} + O(Q^{-4}) \quad (Q \rightarrow \infty).$$

The proof of Theorem 1.2 is divided into three steps, each of independent interest:

1. an explicit formula for the Green function of D_c and the observation that the scaled domains $(D_c)_a$ are again of the same form (Section 2.1);
2. an exact reduction of the extremal problem defining $K_{D_q}^{z^6}(0)$ to the third Taylor coefficient functional on a weighted Bergman space of the unit disk (Section 2.2);
3. a rigorous asymptotic expansion of the associated infinite Gram matrix, valid up to order q^{-4} , which yields the precise size of the convexity defect (Section 2.3).

1.2 Higher order Suita's conjecture

Let Ω be a planar domain in $\mathbb{C}$ bounded by n analytic Jordan curves: $\Gamma_1, \dots, \Gamma_n$, where we set Γ_n to be the boundary of the unbounded component of $\mathbb{C} \setminus \Omega$. According to the solvability of the Dirichlet problem on planar domains, for each $j \in \{1, \dots, n-1\}$ there exists a harmonic function u_j on $\bar{\Omega}$ such that

$$u_j|_{\Gamma_j} \equiv 1, \quad u_j|_{\Gamma_k} \equiv 0 \quad \forall k \in \{1, \dots, n\} \setminus \{j\}.$$

We recall the higher order Suita conjecture in the form in which it is used below. The classical Suita conjecture asserts that for a planar domain Ω with Green function G_Ω and $w \in \Omega$,

$$K_\Omega(w) \geq \frac{c_\Omega(w)^2}{\pi},$$

where

$$c_\Omega(w) := \exp \left(\lim_{z \rightarrow w} (G_\Omega(z, w) - \log |z - w|) \right)$$

is the logarithmic capacity of $\mathbb{C} \setminus \Omega$ with respect to w ; for the unit disk the two sides coincide. Its m -th order analogue, proved in [3], states that for every $m \geq 0$,

$$K_{\Omega}^{(m)}(w) \geq \frac{m!(m+1)!c_{\Omega}(w)^{2m+2}}{\pi},$$

with equality again for the unit disk. In higher dimensions, Blocki–Zwonek [4] proved the general inequality $K_D^H(z) \geq K_{I_D(z)}^H(0)$ for every homogeneous polynomial H , where $I_D(z)$ denotes the Azukawa indicatrix of D at z ; in dimension one, for $H(z) = z^m$, it is exactly the inequality above.

Guan–Sun–Yuan [7] proved a weighted version of the Suita conjecture for higher derivatives, and their result yields the following characterization of the equality locus: for a planar domain bounded by n analytic Jordan curves, the equality in the m -th order Suita conjecture holds at z if and only if

$$(m+1)u_j(z) \in \mathbb{Z}, \quad j = 1, \dots, n-1.$$

Thus, if all $u_j(z)$ are rational, then choosing $m+1$ to be a common denominator of the $u_j(z)$ gives equality at z ; conversely any such equality forces $u_j(z) \in \mathbb{Q}$ for all j . Hence a domain in which, at every point, some triple $(u_1(z), u_2(z), u_3(z))$ fails to be rational has the property that equality in the higher order Suita conjecture never holds.

Recently, in the Annales de l’Institut Fourier, Xu–Zhou [9] constructed, for every $n \geq 3$ and every sufficiently large integer M , a family of smoothly bounded n -connected planar domains on which the equality in the m -th order Suita conjecture fails at every point for each $m = 0, 1, \dots, M$. Our next result complements and strengthens this: it exhibits a four-connected domain on which the equality in the higher order Suita conjecture fails at every point and for every order $m \geq 0$.

THEOREM 1.3. *There exists a planar domain Ω in $\mathbb{C}$ bounded by 4 analytic Jordan curves, such that for the corresponding harmonic functions u_j , $j = 1, 2, 3$, there does not exist $z \in \Omega$ with*

$$u_j(z) \in \mathbb{Q}, \quad j = 1, 2, 3.$$

In particular, at no point of Ω does the equality in the higher order Suita conjecture hold.

We prove Theorem 1.3 by a Baire-category (generic parameter) construction. The same construction works for every connectivity $n \geq 4$; we record this generalization in Subsection 3.1.

1.3 The ξ -reduction fails for $0 < p < 1$

For $p > 0$ and a homogeneous polynomial H of degree k , the higher order p -Bergman kernel $K_D^{H,p}(z)$ and, for a functional $\xi \in \ell_1^{(n)}$, the L^p ξ -Bergman kernel $K_{\xi,D,p}(z)$ are defined in Section 4. Bao–Guan–Sun [10] proved that for every $p \in [1, +\infty)$,

$$K_D^{H,p}(z) = \inf_{\xi \in S_H} K_{\xi,D,p}(z),$$

where S_H is the affine family of functionals whose degree- k part is prescribed by H ; see Theorem 1.7 in [10].

It is natural to ask: whether the identity persists for $0 < p < 1$? Our third counterexample answers this question in the negative. Already in the simplest one-dimensional case the equality fails.

THEOREM 1.4. *Let $\mathbb{D} \subset \mathbb{C}$ be the unit disk, $p = 1/2$, $H(z) = z$ and $z = 0$. Then*

$$K_{\mathbb{D}}^{H,1/2}(0) = \frac{5}{4\pi},$$

while

$$\inf_{\xi \in S_H} K_{\xi, \mathbb{D}, 1/2}(0) > \frac{5}{4\pi}.$$

Consequently the identity

$$K_{\mathbb{D}}^{H,1/2}(0) = \inf_{\xi \in S_H} K_{\xi, \mathbb{D}, 1/2}(0)$$

fails, and the infimum is not attained at the common value.

1.4 Monotonicity fails for $p > 2$

Recall that for $p > 0$, a homogeneous polynomial H of degree k , and a domain $D \subset \mathbb{C}^n$ containing the origin, the higher order p -Bergman kernel $K_D^{H,p}(0)$ is defined in Section 4, and that $D_a := e^{-a}\{G_D(\cdot, 0) < a\}$ for $a \leq 0$. The main theorem of [4] asserts that, for $p = 2$,

$$(-\infty, 0] \ni a \mapsto K_{D_a}^{H,p}(0) \quad \text{is non-decreasing.}$$

In [10] this was extended to all $p \in [1, 2]$: by [10, Theorem 1.7] one has $K_D^{H,p}(z) = \inf_{\xi \in S_H} K_{\xi, D, p}(z)$ for $p \geq 1$, while by [10, Theorem 1.8] the functions

$$(-\infty, 0] \ni a \mapsto \log K_{\xi, \{G_D(\cdot, 0) < a\}, p}(0)$$

are convex and the functions

$$(-\infty, 0] \ni a \mapsto e^{(2n+pk)a} K_{\xi, \{G_D(\cdot, 0) < a\}, p}(0)$$

are non-decreasing for $p \in (0, 2]$; combining the two results, [10, Corollary 1.10] yields the monotonicity of $a \mapsto K_{D_a}^{H,p}(0)$ for every $p \in [1, 2]$. Since the two ingredients have complementary ranges ($p \geq 1$ for the inf-min identity and $p \leq 2$ for the convexity of the ξ -kernels), it is natural to ask whether the monotonicity persists in the two remaining regimes $p \in (0, 1)$ and $p \in (2, +\infty)$.

Our fourth counterexample answers this question in the negative for every $p > 2$. The simplest instance, treated in detail below, uses again the domain D_c and the polynomial $H(z) = z^6$ of Theorem 1.2 and covers $p > 3$; the full range $p > 2$ is then reached through the r -fold symmetric families of Theorem 1.6. The counterexamples were obtained by Codex with Deepseek v4 pro.

THEOREM 1.5. *Let $p > 3$ and $H(z) = z^6$. Then there is a constant $c > 1$ such that, with $D_{1/c} = (D_c)_{-\log c}$,*

$$K_{D_{1/c}}^{z^6, p}(0) > K_{D_c}^{z^6, p}(0).$$

Consequently the function

$$(-\infty, 0] \ni a \mapsto K_{(D_c)_a}^{z^6, p}(0)$$

is not non-decreasing. More precisely, there are constants $0 < c_p < C_p < +\infty$, depending only on p , such that for every $q \geq 8$,

$$c_p q^{-(p-3)} \leq K_{D_q}^{z^6, p}(0) \leq C_p q^{-(p-3)},$$

while

$$\lim_{q \rightarrow 0^+} K_{D_q}^{z^6, p}(0) = \frac{(6!)^p(6p+2)}{2\pi} = K_{\mathbb{D}}^{z^6, p}(0) = K_{I_{D_c}(0)}^{z^6, p}(0),$$

where $\mathbb{D}$ is the unit disk and $I_{D_c}(0)$ is the Azukawa indicatrix of D_c at 0.

THEOREM 1.6. *For every $p > 2$ there is an integer $r > 2/(p-2)$ with the following property. For every integer $m \geq 1$, with $H(z) = z^{rm}$ and*

$$D_q^{(r)} := \{x \in \mathbb{C} : |x|^r < |1 - qx^r|\},$$

there are constants $0 < c_{p,r,m} < C_{p,r,m} < +\infty$, depending only on p, r, m , such that for every $q \geq 8$,

$$c_{p,r,m} q^{-(p-2-2/r)} \leq K_{D_q^{(r)}}^{z^{rm}, p}(0) \leq C_{p,r,m} q^{-(p-2-2/r)},$$

while

$$\lim_{q \rightarrow 0^+} K_{D_q^{(r)}}^{z^{rm}, p}(0) = \frac{(rm)!^p(rmp+2)}{2\pi}.$$

Consequently, for every sufficiently large $c > 1$, with $D_{1/c}^{(r)} = (D_c^{(r)})_{-\frac{1}{r} \log c}$, one has

$$K_{D_{1/c}^{(r)}}^{z^{rm}, p}(0) > K_{D_c^{(r)}}^{z^{rm}, p}(0),$$

and the function $a \mapsto K_{(D_c^{(r)})_a}^{z^{rm}, p}(0)$ is not non-decreasing.

Thus the monotonicity part of the Blocki–Zwonek and Bao–Guan–Sun results has no analogue for any $p > 2$. For the family D_q of Theorem 1.5 the borderline exponent q^{3-p} flips sign exactly at $p = 3$; the r -fold families of Theorem 1.6 push this threshold arbitrarily close to 2 by taking $r > 2/(p-2)$ large. The limiting comparison with the Azukawa indicatrix, however, still holds with equality for all these families. The regime $p \in (0, 1)$ remains open.

1.5 The L^p concavity of minimal integrals fails for $p > 2$

Following [8], let $D \subset \mathbb{C}^n$ be a pseudoconvex domain containing the origin o , let I be an ideal of $\mathcal{O}_o$, and let $(F, o) \in \mathcal{O}_o$. The minimal L^2 integral of D is

$$C_{F,I}(D) := \inf \left\{ \int_D |\tilde{F}|^2 : (\tilde{F} - F, o) \in I, \tilde{F} \in \mathcal{O}(D) \right\}.$$

Guan [8] established a concavity property of the minimal L^2 integrals on the sublevel sets of a plurisubharmonic function: if φ is a negative plurisubharmonic function on D with $\varphi(o) = -\infty$, and

$$G(t) := C_{F, \mathcal{I}(\varphi)_o}(\{\varphi < -t\} \cap D), \quad t \geq 0,$$

where $\mathcal{I}(\varphi)_o$ denotes the multiplier ideal of φ at o (the ideal of germs $f \in \mathcal{O}_o$ with $|f|^2 e^{-\varphi}$ locally integrable near o), then $G(0) < +\infty$ implies that $G(-\log r)$ is concave in $r \in (0, 1]$. This concavity has been a basic tool in the characterization of the equality cases of the optimal L^2 extension theorem and in the higher dimensional generalization of the Suita conjecture.

The monotonicity results of [4] and [10] for the higher order p -Bergman kernels along the Green sublevel sets, recalled in the previous subsection, are the kernel-side counterparts of this structural

property. In their spirit, it is natural to ask whether the L^p analogue of Guan's concavity holds: for $p > 0$ define the minimal L^p integral

$$C_{F,I,p}(D) := \inf \left\{ \int_D |\tilde{F}|^p : (\tilde{F} - F, o) \in I, \tilde{F} \in \mathcal{O}(D) \right\},$$

set

$$G_p(t) := C_{F, \mathcal{I}(\varphi)_{o,p}}(\{\varphi < -t\} \cap D), \quad t \geq 0,$$

and ask whether $G_p(0) < +\infty$ implies that $G_p(-\log r)$ is concave in $r \in (0, 1]$. For $p = 2$ this is exactly Guan's theorem.

Our fifth counterexample answers this question in the negative for every $p > 2$: the r -fold families $D_q^{(r)}$ and the polynomial $H(z) = z^{rm}$ of Theorem 1.6 are transplanted into the minimal L^p integral setting through the multiplier ideal of the Green function. The counterexample was obtained by Codex with Deepseek v4 pro.

THEOREM 1.7. *Let $p > 2$, let $r \geq 1$ and $m \geq 1$ be integers, and set $k := rm$. Fix $c \in (0, 1)$, put*

$$D := D_c^{(r)} = \{x \in \mathbb{C} : |x|^r < |1 - cx^r|\},$$

which is a bounded pseudoconvex domain containing 0, and define

$$\varphi := 2(k+1)G_D(\cdot, 0), \quad F(x) := \frac{x^k}{k!},$$

where $G_D(\cdot, 0)$ is the Green function of D with pole at 0. Then $G_p(0) < +\infty$ and, as $\rho \rightarrow 0^+$,

$$G_p(-\log \rho) \sim \frac{2\pi}{(k!)^p(kp+2)} \rho^\mu, \quad \mu := \frac{pk+2}{2(k+1)} = 1 + \frac{k(p-2)}{2(k+1)} > 1.$$

Consequently $G_p(-\log \rho)$ is not concave in $\rho \in (0, 1]$, and the L^p analogue of Guan's concavity theorem fails for every $p > 2$.

The exponent μ makes the threshold $p = 2$ visible: for $p = 2$ one has $\mu = 1$, and the asymptotic is the linear growth compatible with Guan's concavity; for $0 < p < 2$ one has $\mu < 1$, so the same construction produces no contradiction, and the question remains open in that range. The proof is given in Section 6.

2 Non-convexity of the logarithm of a higher order Bergman kernel

The proof of Theorem 1.2 is divided into three steps, which are treated in the next three subsections.

2.1 The Green function of D_c and the scaling law

Throughout this section we write $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$.

PROPOSITION 2.1. *Let $c > 0$ and*

$$F_c(z) := \frac{z^2}{1 - cz^2}.$$

Then the domain $D_c = \{|F_c| < 1\}$ satisfies $0 \in D_c$, and the mapping

$$F_c : D_c \longrightarrow \mathbb{D} \setminus \{-1/c\}$$

is proper of degree 2, vanishing to order 2 at 0. Consequently the (negative) pluricomplex Green function of D_c with pole at 0 is

$$G_{D_c}(z, 0) = \frac{1}{2} \log |F_c(z)| = \log |z| - \frac{1}{2} \log |1 - cz^2|.$$

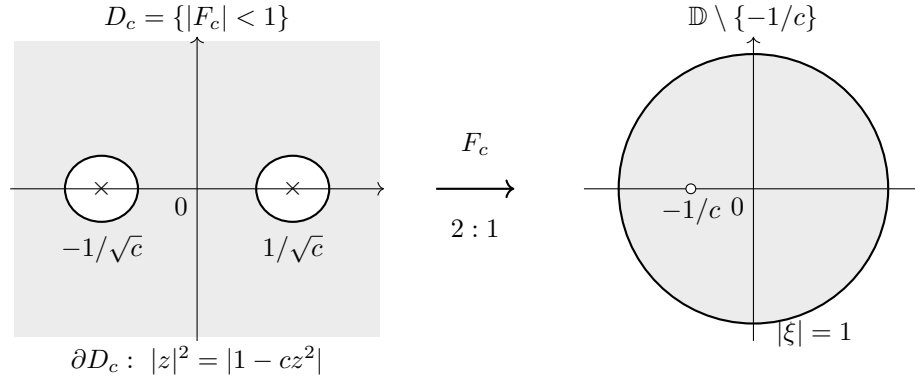


Figure 1: The domain $D_c = \{z \in \mathbb{C} : |z|^2 < |1 - cz^2|\}$ in the z -plane (shaded; the two holes are centered at $\pm 1/\sqrt{c}$ and the boundary is $|z|^2 = |1 - cz^2|$) and its image $F_c(D_c) = \mathbb{D} \setminus \{-1/c\}$ in the ξ -plane. The map F_c is proper of degree 2 with $F_c(0) = 0$ (a double zero) and $F_c(\infty) = -1/c$.

Proof. The identity $D_c = \{|F_c| < 1\}$ is immediate from the definitions. The rational function F_c is holomorphic on D_c : the poles $\pm c^{-1/2}$ are not contained in D_c , since $|F_c| \rightarrow +\infty$ near them. The equation $F_c(z) = \xi$ reads $z^2 = \xi(1 - cz^2)$, i.e. $z^2 = \xi/(1 + c\xi)$, which has two solutions for every $\xi \notin \{-1/c, \infty\}$, so F_c is proper of degree 2 and $F_c(0) = 0$ with $F'_c(0) = 0$, $F''_c(0) = 2 \neq 0$.

For a proper holomorphic map of degree m between planar domains with $f(0) = 0$, the Green functions satisfy $G_D(z, 0) = \frac{1}{m} G_{D'}(f(z), 0)$. Since $\mathbb{D} \setminus \{-1/c\}$ is obtained from $\mathbb{D}$ by removing a polar set, its Green function with pole 0 is $\log |\cdot|$. Taking $m = 2$ gives the formula. $\blacksquare$

For $a \leq 0$ we use the notation of [4]:

$$(D_c)_a := e^{-a} \{G_{D_c}(\cdot, 0) < a\}.$$

PROPOSITION 2.2. *For $a \leq 0$ and $q := ce^{2a}$ we have*

$$(D_c)_a = D_q := \{x \in \mathbb{C} : |x|^2 < |1 - qx^2|\}.$$

Proof. By Proposition 2.1, $G_{D_c}(z, 0) < a$ is equivalent to $|F_c(z)| < e^{2a}$. Writing $z = e^a x$, we have

$$F_c(e^a x) = e^{2a} \frac{x^2}{1 - ce^{2a}x^2} = e^{2a} F_q(x),$$

with $q = ce^{2a}$. Hence $|F_c(e^a x)| < e^{2a}$ is equivalent to $|F_q(x)| < 1$, i.e. $x \in D_q$. Since $(D_c)_a = \{x : e^a x \in \{G_{D_c}(\cdot, 0) < a\}\}$, the claim follows. $\blacksquare$

Since $\log q = \log c + 2a$ is affine in a , the convexity of $a \mapsto \log K_{(D_c)_a}^{z^6}(0)$ is equivalent to the convexity of $s \mapsto \log K_{D_{e^s}}^{z^6}(0)$ in $s = \log q$. We shall therefore study the family D_q , $q > 1$.

2.2 Reduction to a weighted Bergman kernel on the unit disk

For $q > 1$ we abbreviate $K_q := K_{D_q}^{z^6}(0)$. Since D_q is invariant under $x \mapsto -x$, the sixth derivative at 0 only sees the even part of a holomorphic function, and we may restrict to

$$f(x) = g(x^2), \quad g \in \mathcal{O}(U_q),$$

where

$$U_q := \{y \in \mathbb{C} : |y| < |1 - qy|\} = \mathbb{C} \setminus \overline{B\left(\frac{q}{q^2 - 1}, \frac{1}{q^2 - 1}\right)}$$

is the image of D_q under the two-to-one map $y = x^2$.

We record the elementary facts used below.

LEMMA 2.3. *With the notation above, the following hold.*

(i) *For $f(x) = g(x^2)$,*

$$\|f\|_{L^2(D_q)}^2 = \frac{1}{2} \int_{U_q} |g(y)|^2 \frac{dA(y)}{|y|}.$$

(ii) *The conditions $f(0) = f'(0) = \dots = f^{(5)}(0) = 0$ and $f^{(6)}(0) = 1$ are equivalent to*

$$g(y) = b_3 y^3 + O(y^4) \quad (y \rightarrow 0), \quad b_3 = \frac{1}{6!}.$$

(iii) *Under the Möbius transformation*

$$\zeta = \frac{y}{1 - qy}, \quad y = \frac{\zeta}{1 + q\zeta},$$

which maps U_q biholomorphically onto $\mathbb{D} \setminus \{-1/q\}$, the square integrability of $|g|^2/|y|$ forces $g(y(\zeta))$ to vanish at $\zeta = -1/q$. Hence there exists $h \in \mathcal{O}(\mathbb{D})$ with

$$g(y(\zeta)) = (1 + q\zeta)h(\zeta),$$

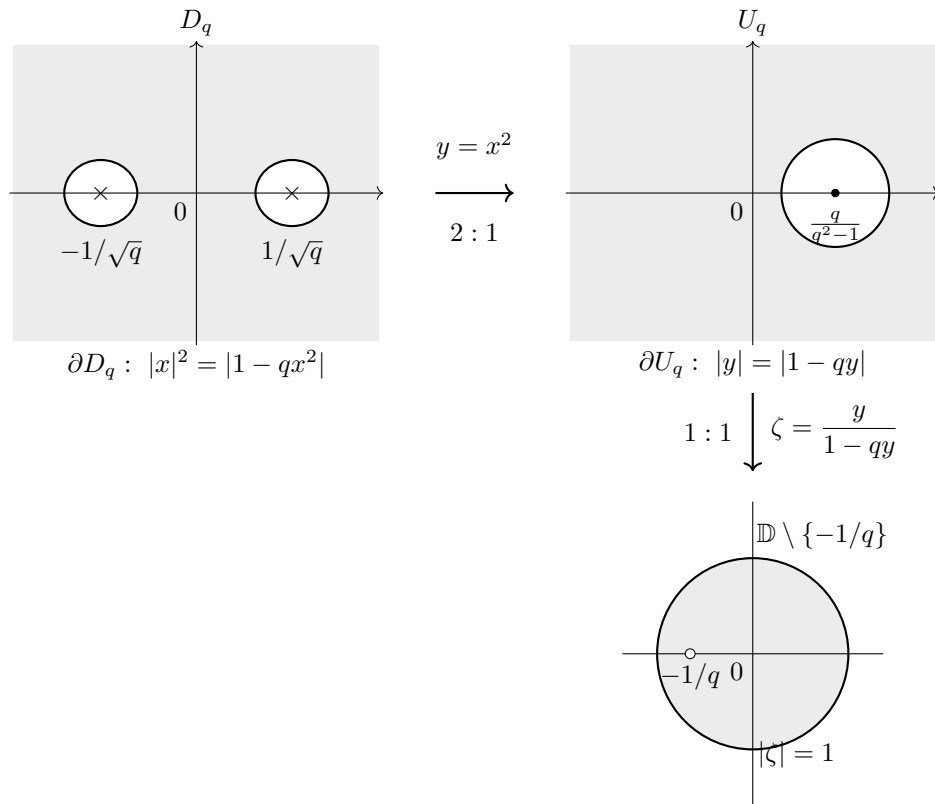


Figure 2: The domain $D_q = \{x \in \mathbb{C} : |x|^2 < |1 - qx^2|\}$ in the x -plane (shaded; the two holes are centered at $\pm\sqrt{q/(q^2 - 1)}$ and contain the poles $\pm 1/\sqrt{q}$), its image $U_q = \{y \in \mathbb{C} : |y| < |1 - qy|\} = \mathbb{C} \setminus \overline{B(q/(q^2 - 1), 1/(q^2 - 1))}$ in the y -plane under the two-to-one map $y = x^2$, and the unit disk $\mathbb{D} \setminus \{-1/q\}$ in the ζ -plane obtained via the Möbius map $\zeta = y/(1 - qy)$ (1 : 1).

and

$$\|f\|_{L^2(D_q)}^2 = \frac{1}{2} \int_{\mathbb{D}} |h(\zeta)|^2 \frac{dA(\zeta)}{|\zeta| |1 + q\zeta|}.$$

(iv) In the variable ζ , the normalization $b_3 = 1/6!$ becomes

$$h(\zeta) = \frac{1}{6!} \zeta^3 + O(\zeta^4).$$

Proof. (i) The map $y = x^2$ is two-to-one from D_q onto U_q , and $dA(y) = 4|x|^2 dA(x) = 4|y| dA(x)$, whence $dA(x) = dA(y)/(4|y|)$.

(ii) Writing $g(y) = \sum_{m \geq 0} g_m y^m$, we have $f(x) = \sum_m g_m x^{2m}$; the first six derivatives vanish at 0 iff $g_0 = g_1 = g_2 = 0$, and $f^{(6)}(0) = 6! g_3$.

(iii) The Möbius map is a biholomorphism between U_q and $\mathbb{D} \setminus \{-1/q\}$ because $y = \zeta/(1 + q\zeta)$ and

$$1 - qy = \frac{1}{1 + q\zeta}, \quad |y| < |1 - qy| \iff |\zeta| < 1.$$

Near $\zeta = -1/q$ one has $y \sim -1/(q^2(\zeta + 1/q)) \rightarrow \infty$. Since g is holomorphic on U_q , including the point at infinity, $g(y(\zeta))$ is holomorphic near $\zeta = -1/q$; if it did not vanish there, then $|g|^2/|y| \sim |g(-1/q)|^2 |1 + q\zeta|^3$ would not be integrable near $\zeta = -1/q$, contradicting the finiteness of $\|f\|^2$. Hence $g(y(\zeta))$ vanishes at $\zeta = -1/q$, so the quotient $h = g(y(\zeta))/(1 + q\zeta)$ is holomorphic on $\mathbb{D}$. The change of variables $y = \zeta/(1 + q\zeta)$ has Jacobian $dA(y) = |1 + q\zeta|^{-4} dA(\zeta)$, and $|y|^{-1} = |1 + q\zeta|/|\zeta|$, so

$$\frac{1}{2} \int_{U_q} |g|^2 \frac{dA(y)}{|y|} = \frac{1}{2} \int_{\mathbb{D}} |h|^2 \frac{dA(\zeta)}{|\zeta| |1 + q\zeta|}.$$

(iv) From $y = \zeta + O(\zeta^2)$ we get $g(y(\zeta)) = g_3 \zeta^3 + O(\zeta^4)$; dividing by $(1 + q\zeta)$ does not change the leading term. ■

We now introduce the infinite Gram matrix. For $i, j \geq 3$ set

$$M_{ij}(q) := \int_{\mathbb{D}} \frac{\zeta^i \bar{\zeta}^j}{|\zeta| |1 + q\zeta|} dA(\zeta),$$

and let $K_3(q) := (M(q)^{-1})_{33}$ be the $(3,3)$ -entry of the inverse of the infinite matrix $M(q)$ (which is positive definite on the space of holomorphic functions h with $\int |h|^2/(|\zeta| |1 + q\zeta|) dA < \infty$).

PROPOSITION 2.4. *For every $q > 1$,*

$$K_q = K_{D_q}^{z^6}(0) = 2(6!)^2 K_3(q).$$

Proof. By definition,

$$K_q^{-1} = \min \left\{ \|f\|_{L^2(D_q)}^2 : f \in \mathcal{O}(D_q), f^{(j)}(0) = 0 \ (0 \leq j < 6), f^{(6)}(0) = 1 \right\}.$$

By Lemma 2.3 this equals

$$\min \left\{ \frac{1}{2} \int_{\mathbb{D}} |h|^2 \frac{dA}{|\zeta| |1 + q\zeta|} : h \in \mathcal{O}(\mathbb{D}), h(\zeta) = \frac{1}{6!} \zeta^3 + O(\zeta^4) \right\}.$$

Writing $h(\zeta) = \frac{1}{6!}\zeta^3 + \sum_{m \geq 4} c_m \zeta^m$, the minimum over the higher coefficients is

$$\frac{1}{2(6!)^2} (M(q)^{-1})_{33},$$

and taking the reciprocal gives the claim. ■

2.3 Asymptotics of the Gram matrix

Set $\varepsilon := 1/q$ and $A(\varepsilon) := qM(q)$, so that

$$A_{ij}(\varepsilon) = \int_{\mathbb{D}} \frac{\zeta^i \bar{\zeta}^j}{|\zeta| |\zeta + \varepsilon|} dA(\zeta), \quad i, j \geq 3.$$

We write $D = \text{diag}(\pi/3, \pi/4, \pi/5, \dots)$, and for the off-diagonal terms we use the convention that $B_{i,i+1} = B_{i+1,i}$ for $i \geq 3$.

PROPOSITION 2.5. *The matrix function $A(\varepsilon)$ admits the expansion*

$$A(\varepsilon) = D + \varepsilon B + \varepsilon^2 C + \varepsilon^3 E + O(\varepsilon^4)$$

in the sense of bounded quadratic forms on the Hilbert space $\{h = \zeta^3 v : v \in L_h^2(\mathbb{D})\}$ with norm $\int |h|^2 |\zeta|^{-2} dA$. The nonzero leading terms are

$$D_{ii} = \frac{\pi}{i}, \quad B_{i,i+1} = B_{i+1,i} = -\frac{\pi}{2i}, \quad C_{ii} = \frac{\pi}{4(i-1)}, \quad i \geq 3.$$

Consequently, for $\varepsilon > 0$ sufficiently small,

$$(A(\varepsilon)^{-1})_{33} = \frac{3}{\pi} - \frac{1}{8\pi} \varepsilon^2 + O(\varepsilon^4).$$

Proof. The expansion of the kernel in the outer region $|\zeta| \geq 2\varepsilon$,

$$\frac{1}{|\zeta + \varepsilon|} = \frac{1}{|\zeta|} \left(1 + \frac{2\varepsilon \operatorname{Re} \zeta}{|\zeta|^2} + \frac{\varepsilon^2}{|\zeta|^2} \right)^{-1/2} = \frac{1}{|\zeta|} - \frac{\varepsilon \operatorname{Re} \zeta}{|\zeta|^3} + \varepsilon^2 \frac{3(\operatorname{Re} \zeta)^2 - |\zeta|^2}{2|\zeta|^5} + O\left(\frac{\varepsilon^3}{|\zeta|^4}\right),$$

together with the elementary bound in the inner disk $|\zeta| < 2\varepsilon$ (where $\zeta = \varepsilon s$, so that all quadratic forms under consideration are $O(\varepsilon^6)$ after testing against $h = \zeta^3 v$), justifies the expansion in the sense of bounded quadratic forms; the remainder is controlled by

$$C\varepsilon^4 \int_{\mathbb{D}} |v|^2 dA \leq C'\varepsilon^4 \int_{\mathbb{D}} |h|^2 |\zeta|^{-2} dA.$$

The angular Fourier integrals give

$$\int_0^{2\pi} e^{id\theta} d\theta = 2\pi \delta_{d0}, \quad \int_0^{2\pi} \cos \theta e^{i\theta} d\theta = \pi, \quad \int_0^{2\pi} (3 \cos^2 \theta - 1) d\theta = \pi,$$

which yield, after the radial integration,

$$D_{ii} = \int_0^1 \rho^{2i} \frac{2\pi}{\rho} d\rho = \frac{\pi}{i},$$

$$B_{i,i+1} = - \int_0^1 \rho^{2i+1} \frac{\pi}{\rho^2} d\rho = -\frac{\pi}{2i},$$

$$C_{ii} = \int_0^1 \rho^{2i} \frac{\pi}{2\rho^3} d\rho = \frac{\pi}{4(i-1)}.$$

The first-order off-diagonal entries of B with $|i-j| \neq 1$ vanish by angular orthogonality.

For the inverse, we use the Neumann expansion

$$A^{-1} = D^{-1} - \varepsilon D^{-1} B D^{-1} + \varepsilon^2 (D^{-1} B D^{-1} B D^{-1} - D^{-1} C D^{-1}) + O(\varepsilon^3).$$

Since D is diagonal, the second-order contribution to the $(3,3)$ -entry receives only the diagonal term C_{33} and the unique off-diagonal path $3 \rightarrow 4 \rightarrow 3$, so that

$$\begin{aligned} (A^{-1})_{33} &= \frac{3}{\pi} + \varepsilon^2 \left(\frac{B_{34}^2}{D_{33}^2 D_{44}} - \frac{C_{33}}{D_{33}^2} \right) + O(\varepsilon^4) \\ &= \frac{3}{\pi} + \varepsilon^2 \left(\frac{(\pi/6)^2}{(\pi/3)^2 (\pi/4)} - \frac{\pi/8}{(\pi/3)^2} \right) + O(\varepsilon^4) \\ &= \frac{3}{\pi} - \frac{1}{8\pi} \varepsilon^2 + O(\varepsilon^4). \end{aligned}$$

There is no third-order term: the transformation $\zeta \mapsto -\zeta$ unitarily identifies $A(\varepsilon)$ with $A(-\varepsilon)$, while the coefficient functional is unchanged, so the expansion of $(A^{-1})_{33}$ contains only even powers. ■

Combining Proposition 2.4 with Proposition 2.5 gives the asymptotic law for the kernel.

COROLLARY 2.6. *As $q \rightarrow \infty$,*

$$K_3(q) = q(A(q^{-1})^{-1})_{33} = \frac{3q}{\pi} \left(1 - \frac{1}{24q^2} + O(q^{-4}) \right),$$

and therefore, with $C_0 := \log(2(6!)^2 \cdot 3/\pi)$,

$$\log K_{D_q}^{z^6}(0) = C_0 + \log q - \frac{1}{24q^2} + O(q^{-4}).$$

2.4 Proof of the main theorem

Proof of Theorem 1.2. Let $Q > 1$ and $c = 2Q$. By Proposition 2.2, the three values

$$a_- = -\log 2, \quad a_0 = -\frac{1}{2} \log 2, \quad a_+ = 0$$

correspond to

$$q_- = \frac{Q}{2}, \quad q_0 = Q, \quad q_+ = 2Q.$$

Since $a_0 = (a_- + a_+)/2$ and $\log q = \log c + 2a$ is affine in a , it suffices to prove

$$\log K_{D_Q}^{z^6}(0) - \frac{1}{2} \left(\log K_{D_{Q/2}}^{z^6}(0) + \log K_{D_{2Q}}^{z^6}(0) \right) > 0$$

for Q sufficiently large. By Corollary 2.6, the terms C_0 cancel, and so do the linear terms $\log q$:

$$\log Q - \frac{1}{2}(\log(Q/2) + \log(2Q)) = 0.$$

The remaining contribution is

$$-\frac{1}{24}\left(\frac{1}{Q^2} - \frac{1}{2}\left(\frac{4}{Q^2} + \frac{1}{4Q^2}\right)\right) + O(Q^{-4}) = \frac{3}{64Q^2} + O(Q^{-4}) > 0$$

for Q large enough. This proves the strict midpoint inequality, hence the failure of convexity. ■

REMARK 2.7 (A different convexity question). We recall that Blocki and Zwonek also conjectured in [3] the convexity of the area function $t \mapsto \log \lambda(\{G_{\Omega,w} < t\})$; counterexamples on annuli were found numerically in [1] and proved in full generality in [3]. That conjecture concerns the volume of sublevel sets and is unrelated to Question 1.1, which concerns the Bergman kernel.

3 A four-connected domain with no point of equality in the higher order Suita conjecture

Fix three pairwise distinct points $c_1, c_2, c_3 \in \mathbb{D}$ and choose $\rho_0 > 0$ so small that the closed disks $\overline{\mathbb{D}(c_j, 2\rho_0)}$ are pairwise disjoint and contained in $\mathbb{D}$. Let

$$B := \left(\frac{1}{2}\rho_0, \rho_0\right) \times \left(\frac{1}{2}\rho_0, \rho_0\right) \times \left(\frac{1}{2}\rho_0, \rho_0\right) \subset \mathbb{R}^3,$$

where $(\frac{1}{2}\rho_0, \rho_0)$ denotes the open interval in $\mathbb{R}$. For $\mathbf{r} = (r_1, r_2, r_3) \in B$ put

$$D_j(\mathbf{r}) := \overline{\mathbb{D}(c_j, r_j)}, \quad \Gamma_j(\mathbf{r}) := \partial D_j(\mathbf{r}), \quad j = 1, 2, 3, \quad \Gamma_4 := \partial \mathbb{D},$$

and

$$\Omega(\mathbf{r}) := \mathbb{D} \setminus \left(D_1(\mathbf{r}) \cup D_2(\mathbf{r}) \cup D_3(\mathbf{r}) \right).$$

Then $\Omega(\mathbf{r})$ is bounded by the four pairwise disjoint analytic Jordan curves $\Gamma_1(\mathbf{r}), \Gamma_2(\mathbf{r}), \Gamma_3(\mathbf{r}), \Gamma_4$ (see Figure 3).

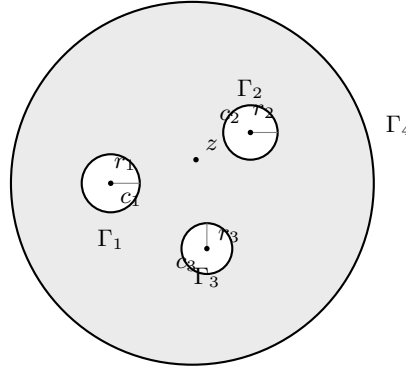


Figure 3: The four-connected domain $\Omega(\mathbf{r}) = \mathbb{D} \setminus (D_1(\mathbf{r}) \cup D_2(\mathbf{r}) \cup D_3(\mathbf{r}))$ with boundary components $\Gamma_1, \Gamma_2, \Gamma_3$ and Γ_4 .

For $j = 1, 2, 3$ let $u_j(\cdot; \mathbf{r})$ be the harmonic function on $\Omega(\mathbf{r})$ with boundary values 1 on $\Gamma_j(\mathbf{r})$ and 0 on the other three curves, and set $u_4 := 1 - u_1 - u_2 - u_3$. Basic facts on harmonic measure used below can be found in [6].

LEMMA 3.1. *For every $z \in \Omega(\mathbf{r})$ and $j, k \in \{1, 2, 3\}$,*

$$\frac{\partial u_j}{\partial r_j}(z; \mathbf{r}) > 0, \quad \frac{\partial u_j}{\partial r_k}(z; \mathbf{r}) < 0 \quad (j \neq k), \quad \frac{\partial u_4}{\partial r_k}(z; \mathbf{r}) < 0. \quad (3.1)$$

Proof. The functions u_j are real-analytic in $(z, \mathbf{r})$; this follows from the standard analytic dependence of solutions of the Dirichlet problem on analytic boundary data [11]. The first inequality in (3.1) is the usual monotonicity of harmonic measure under enlargement of the corresponding boundary component (Figure 4); it follows from the maximum principle, since on the enlarged domain the old solution is < 1 on the new boundary and $= 0$ on all other boundary components.

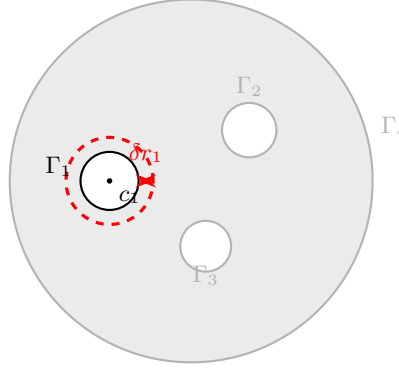


Figure 4: Enlargement of the boundary component $\Gamma_1(\mathbf{r})$ by $\delta r_1 > 0$: the dashed circle is the new boundary, and the red arrow indicates the outward normal displacement used in the variational formula (3.2).

For $k \neq j$ (and for u_4 with any k) we use Hadamard's variational formula. Let $G_{\mathbf{r}}(w, z)$ be the Green function of $\Omega(\mathbf{r})$ and let n denote the outward unit normal. Up to a fixed positive multiplicative constant,

$$\frac{\partial u_j}{\partial r_k}(z; \mathbf{r}) = \int_{\Gamma_k(\mathbf{r})} \frac{\partial u_j}{\partial n}(w) \frac{\partial G_{\mathbf{r}}}{\partial n}(w, z) ds(w). \quad (3.2)$$

For an inner boundary $\Gamma_k(\mathbf{r})$ one has $\partial G_{\mathbf{r}}/\partial n > 0$ on $\Gamma_k(\mathbf{r})$. By Hopf's lemma, $\partial u_j/\partial n$ on $\Gamma_k(\mathbf{r})$ is strictly positive if $j = k$ and strictly negative if $j \neq k$; similarly $\partial u_4/\partial n < 0$ on $\Gamma_k(\mathbf{r})$. Hence (3.1) follows from (3.2). $\blacksquare$

For $z \in \Omega(\mathbf{r})$ put

$$A(z; \mathbf{r}) := \left(\frac{\partial u_j}{\partial r_k}(z; \mathbf{r}) \right)_{1 \leq j, k \leq 3}.$$

LEMMA 3.2. *For every $\mathbf{r} \in B$ and every $z \in \Omega(\mathbf{r})$ the matrix $A(z; \mathbf{r})$ is invertible.*

Proof. By (3.1), $A_{jk} < 0$ for $j \neq k$, and

$$(A^T \mathbf{1})_j = \sum_{i=1}^3 \frac{\partial u_i}{\partial r_j} = \frac{\partial}{\partial r_j} (u_1 + u_2 + u_3) = -\frac{\partial u_4}{\partial r_j} > 0.$$

Thus A^T has positive diagonal, nonpositive off-diagonal entries and positive row sums. If $A^T \mathbf{x} = 0$ for some $\mathbf{x} \neq 0$, choose an index i at which $x_i = \max_j x_j > 0$; if all $x_j \leq 0$, choose instead an index i at which $x_i = \min_j x_j < 0$. In the first case, since $A_{ik}^T \leq 0$ for $k \neq i$,

$$0 = (A^T \mathbf{x})_i = A_{ii}^T x_i + \sum_{k \neq i} A_{ik}^T x_k \geq A_{ii}^T x_i + \sum_{k \neq i} A_{ik}^T x_i = (A^T \mathbf{1})_i x_i > 0,$$

a contradiction. The second case is analogous with all inequalities reversed. Hence A^T , and therefore A , is nonsingular. $\blacksquare$

Proof of Theorem 1.3. Let

$$X := \{(z, \mathbf{r}) \in \mathbb{C} \times B : z \in \Omega(\mathbf{r})\},$$

an open subset of $\mathbb{R}^2 \times \mathbb{R}^3 \cong \mathbb{R}^5$, and define

$$F : X \rightarrow \mathbb{R}^3, \quad F(z, \mathbf{r}) := (u_1(z; \mathbf{r}), u_2(z; \mathbf{r}), u_3(z; \mathbf{r})).$$

By Lemma 3.1, F is real-analytic. Its derivative contains the invertible 3×3 block $D_{\mathbf{r}}F = A(z; \mathbf{r})$; hence F is a submersion at every point of X .

For $\mathbf{q} = (q_1, q_2, q_3) \in \mathbb{Q}^3$ set $S_{\mathbf{q}} := F^{-1}(\mathbf{q})$. By the implicit function theorem, $S_{\mathbf{q}}$ is a real-analytic submanifold of X of dimension $\dim_{\mathbb{R}} X - 3 = 2$ (possibly empty). Let $\pi_{\mathbf{q}} : S_{\mathbf{q}} \rightarrow B$ be the projection onto the parameter box. Each connected component of $S_{\mathbf{q}}$ is a two-dimensional C^1 manifold. Since $\dim_{\mathbb{R}} B = 3$, Sard's theorem shows that its image under $\pi_{\mathbf{q}}$ has empty interior (indeed Lebesgue measure zero). A second countable manifold has at most countably many connected components, so $\pi_{\mathbf{q}}(S_{\mathbf{q}})$ is a countable union of null sets, hence a null set. Consequently

$$E := \bigcup_{\mathbf{q} \in \mathbb{Q}^3} \pi_{\mathbf{q}}(S_{\mathbf{q}})$$

is a countable union of null sets, and $B \setminus E$ is nonempty (in fact of full measure).

Choose $\mathbf{r} \in B \setminus E$. We claim that $\Omega := \Omega(\mathbf{r})$ has the required property. Indeed, if for some $z \in \Omega$ all three numbers $u_j(z; \mathbf{r})$ were rational, then with $\mathbf{q} := (u_1(z; \mathbf{r}), u_2(z; \mathbf{r}), u_3(z; \mathbf{r})) \in \mathbb{Q}^3$ we would have $(z, \mathbf{r}) \in S_{\mathbf{q}}$ and hence $\mathbf{r} \in E$, a contradiction. This completes the proof. $\blacksquare$

3.1 Generalization to arbitrary connectivity

The construction of the preceding part extends verbatim to every $n \geq 4$.

THEOREM 3.3. *For every integer $n \geq 4$ there exists a planar domain $\Omega \subset \mathbb{C}$ bounded by n analytic Jordan curves such that, for the corresponding harmonic functions $u_1, \dots, u_{n-1}$, there does not exist $z \in \Omega$ with*

$$u_j(z) \in \mathbb{Q}, \quad j = 1, \dots, n-1.$$

In particular, at no point of Ω does the equality in the higher order Suita conjecture hold.

Proof. Fix $n \geq 4$ and set $m := n - 1$. Choose m pairwise distinct points $c_1, \dots, c_m \in \mathbb{D}$ and $\rho_0 > 0$ so small that the closed disks $\overline{\mathbb{D}(c_j, 2\rho_0)}$ are pairwise disjoint and contained in $\mathbb{D}$. Let $B := (\rho_0/2, \rho_0)^m$. For $\mathbf{r} = (r_1, \dots, r_m) \in B$ define

$$D_j(\mathbf{r}) := \overline{\mathbb{D}(c_j, r_j)}, \quad \Gamma_j(\mathbf{r}) := \partial D_j(\mathbf{r}) \quad (1 \leq j \leq m), \quad \Gamma_{m+1} := \partial \mathbb{D},$$

and

$$\Omega(\mathbf{r}) := \mathbb{D} \setminus \bigcup_{j=1}^m D_j(\mathbf{r}).$$

Let $u_j(\cdot; \mathbf{r})$ be the harmonic function on $\Omega(\mathbf{r})$ with boundary values 1 on $\Gamma_j(\mathbf{r})$ and 0 on the other curves, $1 \leq j \leq m$, and put $u_{m+1} := 1 - \sum_{j=1}^m u_j$.

The variational argument in Lemma 3.1 applies unchanged with m in place of 3: for all $j, k \in \{1, \dots, m\}$,

$$\frac{\partial u_j}{\partial r_j}(z; \mathbf{r}) > 0, \quad \frac{\partial u_j}{\partial r_k}(z; \mathbf{r}) < 0 \quad (j \neq k), \quad \frac{\partial u_{m+1}}{\partial r_k}(z; \mathbf{r}) < 0.$$

Indeed, the off-diagonal and u_{m+1} estimates follow from Hadamard's formula (3.2) and Hopf's lemma exactly as in the proof of Lemma 3.1.

Let $A(z; \mathbf{r}) := (\partial u_j / \partial r_k)_{1 \leq j, k \leq m}$. As in Lemma 3.2,

$$(A^T \mathbf{1})_j = \sum_{i=1}^m \frac{\partial u_i}{\partial r_j} = \frac{\partial}{\partial r_j} \left(\sum_{i=1}^m u_i \right) = -\frac{\partial u_{m+1}}{\partial r_j} > 0.$$

Thus A^T has positive diagonal, nonpositive off-diagonal entries and positive row sums; by the same maximum-entry argument as in Lemma 3.2, A^T , and hence A , is nonsingular.

Define

$$X := \{(z, \mathbf{r}) \in \mathbb{C} \times B : z \in \Omega(\mathbf{r})\}, \quad F : X \rightarrow \mathbb{R}^m, \quad F(z, \mathbf{r}) := (u_1(z; \mathbf{r}), \dots, u_m(z; \mathbf{r})).$$

Then F is real-analytic and a submersion, since $D_{\mathbf{r}} F = A$ is invertible. For every $\mathbf{q} \in \mathbb{Q}^m$, the set $S_{\mathbf{q}} := F^{-1}(\mathbf{q})$ is a real-analytic submanifold of X of dimension

$$\dim_{\mathbb{R}} X - m = (2 + m) - m = 2.$$

Let $\pi_{\mathbf{q}} : S_{\mathbf{q}} \rightarrow B$ be the projection onto the parameter box. Since $\dim_{\mathbb{R}} B = m = n - 1 \geq 3$, Sard's theorem shows that $\pi_{\mathbf{q}}(S_{\mathbf{q}})$ is a null set: every connected component is a smooth image of a two-dimensional manifold, and there are at most countably many components. Therefore

$$E := \bigcup_{\mathbf{q} \in \mathbb{Q}^m} \pi_{\mathbf{q}}(S_{\mathbf{q}})$$

is a countable union of null sets, and $B \setminus E$ is nonempty.

Choose $\mathbf{r} \in B \setminus E$. If for some $z \in \Omega(\mathbf{r})$ all $u_j(z; \mathbf{r})$ were rational, then $(z, \mathbf{r}) \in S_{\mathbf{q}}$ for $\mathbf{q} := F(z, \mathbf{r}) \in \mathbb{Q}^m$, so $\mathbf{r} \in E$, a contradiction. Hence $\Omega := \Omega(\mathbf{r})$ satisfies the conclusion of the theorem. $\blacksquare$

4 The ξ -reduction fails for $0 < p < 1$

Let $D \subset \mathbb{C}^n$ be a domain, $p > 0$, and let

$$H(z) = \sum_{|\alpha|=k} a_\alpha z^\alpha$$

be a homogeneous polynomial of degree k . Following [10], denote by

$$\ell_1^{(n)} := \left\{ \xi = (\xi_\alpha)_{\alpha \in \mathbb{N}^n} : \xi_\alpha \in \mathbb{C}, \sum_{\alpha \in \mathbb{N}^n} |\xi_\alpha| \rho^{|\alpha|} < +\infty \text{ for every } \rho > 0 \right\}$$

the space of functionals on holomorphic germs. For $\xi \in \ell_1^{(n)}$ and $f \in \mathcal{O}(D)$, the pairing is defined by

$$(\xi \cdot f)(z) := \sum_{\alpha \in \mathbb{N}^n} \xi_\alpha \frac{D^\alpha f(z)}{\alpha!}.$$

For the homogeneous polynomial H of degree k above, let S_H denote the affine family of all $\xi \in \ell_1^{(n)}$ such that

$$\xi_\alpha = \begin{cases} a_\alpha \alpha!, & |\alpha| = k, \\ 0, & |\alpha| > k, \end{cases}$$

the entries with $|\alpha| < k$ being free. With this notation, the higher order p -Bergman kernel and the L^p ξ -Bergman kernel are

$$K_D^{H,p}(z) := \sup \left\{ |P_H(f)(z)|^p : f \in A^p(D), f^{(j)}(z) = 0 \ (0 \leq j < k), \|f\|_{L^p(D)} \leq 1 \right\},$$

$$K_{\xi,D,p}(z) := \sup \left\{ |(\xi \cdot f)(z)|^p : f \in A^p(D), \|f\|_{L^p(D)} \leq 1 \right\}.$$

For a positive homogeneous functional, equivalently,

$$K_{\xi,D,p}(z) = \frac{1}{\inf \left\{ \|f\|_{L^p(D)}^p : f \in A^p(D), (\xi \cdot f)(z) = 1 \right\}}.$$

We answer the question whether the identity in Theorem 1.7 of [10] extends to $0 < p < 1$ in the negative. Already the one-dimensional model $D = \mathbb{D}$, $H(z) = z$, $p = 1/2$ gives a counterexample. For $H(z) = z$ the set S_H consists of all sequences

$$\xi = (\xi_0, 1, 0, 0, \dots), \quad \xi_0 \in \mathbb{C}.$$

Proof of Theorem 1.4. We first compute $K_{\mathbb{D}}^{H,1/2}(0)$. Let $f \in \mathcal{O}(\mathbb{D})$ satisfy $f(0) = 0$ and $f'(0) = 1$, and write $f(z) = zg(z)$ with $g(0) = 1$. Since $|g|^{1/2}$ is subharmonic, for every $r \in (0, 1)$,

$$\frac{1}{2\pi} \int_0^{2\pi} |g(re^{i\theta})|^{1/2} d\theta \geq |g(0)|^{1/2} = 1.$$

Hence

$$\int_{\mathbb{D}} |f|^{1/2} dA = \int_0^1 r^{1/2} r \left(\int_0^{2\pi} |g(re^{i\theta})|^{1/2} d\theta \right) dr \geq 2\pi \int_0^1 r^{3/2} dr = \frac{4\pi}{5},$$

with equality for $g \equiv 1$, i.e. $f(z) = z$. Therefore

$$K_{\mathbb{D}}^{H,1/2}(0) = \frac{1}{4\pi/5} = \frac{5}{4\pi}.$$

Next, fix $\alpha = 3/2$ and set

$$M(\alpha) := \frac{1}{\pi} \int_{\mathbb{D}} |1 + \alpha z| dA(z).$$

We need the elementary estimate

$$\frac{M(\alpha)}{\sqrt{2\alpha}} < \frac{4}{5}. \quad (4.3)$$

For $0 \leq x \leq 1$ the fourth Taylor polynomial of $\sqrt{1-x}$ gives

$$\sqrt{1-x} \leq 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} - \frac{5x^4}{128}, \quad (4.4)$$

because all remaining Taylor coefficients of $\sqrt{1-x}$ are negative. Write $t = \alpha r$ and

$$\kappa := \frac{4t}{(1+t)^2}.$$

Then

$$|1 + te^{i\theta}| = (1+t) \sqrt{1 - \kappa \sin^2 \frac{\theta}{2}}.$$

Using (4.4) and the averages

$$\frac{1}{2\pi} \int_0^{2\pi} \sin^2 \frac{\theta}{2} d\theta = \frac{1}{2}, \quad \frac{1}{2\pi} \int_0^{2\pi} \sin^4 \frac{\theta}{2} d\theta = \frac{3}{8}, \quad \frac{1}{2\pi} \int_0^{2\pi} \sin^6 \frac{\theta}{2} d\theta = \frac{5}{16}, \quad \frac{1}{2\pi} \int_0^{2\pi} \sin^8 \frac{\theta}{2} d\theta = \frac{35}{128},$$

we obtain

$$\frac{1}{2\pi} \int_0^{2\pi} |1 + te^{i\theta}| d\theta \leq (1+t) \left(1 - \frac{\kappa}{4} - \frac{3\kappa^2}{64} - \frac{5\kappa^3}{256} - \frac{175\kappa^4}{16384} \right).$$

Consequently

$$M(\alpha) \leq \int_0^1 2r(1 + \alpha r) \left(1 - \frac{\kappa}{4} - \frac{3\kappa^2}{64} - \frac{5\kappa^3}{256} - \frac{175\kappa^4}{16384} \right) dr.$$

For $\alpha = 3/2$ the integral on the right is

$$\frac{13771}{10000}.$$

Since

$$\left(\frac{13771}{10000} \right)^2 = \frac{189640441}{100000000} < \frac{48}{25} = \left(\frac{4}{5} \sqrt{3} \right)^2,$$

we have

$$\frac{13771}{10000} < \frac{4}{5} \sqrt{3} = \frac{4}{5} \sqrt{2\alpha},$$

which proves (4.3).

Now fix any $c \in \mathbb{C}$. Choose a unit complex number $e^{i\theta}$ such that c and $2\alpha e^{i\theta}$ have the same argument (take $e^{i\theta} = 1$ if $c = 0$), so that $c + 2\alpha e^{i\theta}$ has modulus $|c| + 2\alpha$. Choose a unit complex number $e^{i\psi}$ with $e^{i\psi}(c + 2\alpha e^{i\theta}) = |c| + 2\alpha$, and define

$$f_{c,\theta}(z) := e^{i\psi} \frac{(1 + \alpha e^{i\theta} z)^2}{|c| + 2\alpha}.$$

Then $f_{c,\theta}$ is a polynomial, and

$$f_{c,\theta}(0) = \frac{e^{i\psi}}{|c| + 2\alpha}, \quad f'_{c,\theta}(0) = \frac{2\alpha e^{i\theta} e^{i\psi}}{|c| + 2\alpha},$$

so that for $\xi_c = (c, 1, 0, 0, \dots) \in S_H$,

$$(\xi_c \cdot f_{c,\theta})(0) = c f_{c,\theta}(0) + f'_{c,\theta}(0) = \frac{e^{i\psi}(c + 2\alpha e^{i\theta})}{|c| + 2\alpha} = 1.$$

Moreover, by the rotation invariance of dA ,

$$\int_{\mathbb{D}} |f_{c,\theta}|^{1/2} dA = \frac{1}{\sqrt{|c| + 2\alpha}} \int_{\mathbb{D}} |1 + \alpha e^{i\theta} z| dA = \frac{\pi M(\alpha)}{\sqrt{|c| + 2\alpha}} \leq \frac{\pi M(\alpha)}{\sqrt{2\alpha}} < \frac{4\pi}{5},$$

since $|c| + 2\alpha \geq 2\alpha$. By the equivalent form of $K_{\xi,D,p}$,

$$K_{\xi_c, \mathbb{D}, 1/2}(0) \geq \frac{1}{\int_{\mathbb{D}} |f_{c,\theta}|^{1/2} dA} > \frac{5}{4\pi} = K_{\mathbb{D}}^{H, 1/2}(0).$$

This holds for every $c \in \mathbb{C}$, i.e. for every $\xi \in S_H$.

Thus for every $\xi \in S_H$,

$$K_{\xi, \mathbb{D}, 1/2}(0) > \frac{5}{4\pi},$$

and hence

$$\inf_{\xi \in S_H} K_{\xi, \mathbb{D}, 1/2}(0) > \frac{5}{4\pi} = K_{\mathbb{D}}^{H, 1/2}(0),$$

which proves the theorem. ■

4.1 The threshold $p_* = 2(\sqrt{2} - 1)$ for the counterexample

The counterexample in Theorem 1.4 is not an isolated phenomenon: the second variation at the extremal $f(z) = z$ shows that the identity

$$K_{\mathbb{D}}^{H,p}(0) = \inf_{\xi \in S_H} K_{\xi, \mathbb{D}, p}(0) \tag{4.5}$$

fails for every $0 < p < p_*$, where

$$p_* := 2(\sqrt{2} - 1).$$

For $p > p_*$ the same computation shows that z is a strict local minimizer in the class $f'(0) = 1$, and at $p = p_*$ the second variation degenerates along a single explicit direction; numerically, z remains the global minimizer for every $p \geq p_*$ in this example.

Since $H(z) = z$, every $\xi \in S_H$ has the form $\xi_c = (c, 1, 0, 0, \dots)$, $c \in \mathbb{C}$, and

$$K_{\xi_c, \mathbb{D}, p}(0) = \frac{1}{m_{\xi_c}(p)}, \quad m_{\xi_c}(p) := \inf \left\{ \int_{\mathbb{D}} |f|^p dA : f \in \mathcal{O}(\mathbb{D}), cf(0) + f'(0) = 1 \right\}.$$

The function $f(z) = z$ is admissible for every ξ_c , and $\int_{\mathbb{D}} |z|^p dA = 2\pi/(p+2)$; hence $m_{\xi_c}(p) \leq 2\pi/(p+2)$ for all c .

THEOREM 4.1. *Let $p_* := 2(\sqrt{2} - 1)$. For every $0 < p < p_*$, with $D = \mathbb{D}$, $H(z) = z$ and $z = 0$,*

$$\inf_{\xi \in S_H} K_{\xi, \mathbb{D}, p}(0) > K_{\mathbb{D}}^{H, p}(0);$$

that is, the identity (4.5) fails for every $p \in (0, p_)$. In particular, Theorem 4.1 contains the case $p = 1/2$ of Theorem 1.4.*

Proof. By the same argument as in the proof of Theorem 1.4, with p in place of $1/2$,

$$K_{\mathbb{D}}^{H, p}(0) = \frac{p+2}{2\pi}.$$

It suffices to show that $m_{\xi_0}(p) < 2\pi/(p+2)$. For $\lambda > 0$ and $\varepsilon \in \mathbb{R}$ consider

$$f_\varepsilon(z) := z + \varepsilon(1 + \lambda z^2),$$

so that $f'_\varepsilon(0) = 1$ and $f_0 = z$. Write $J_p(\varepsilon) := \int_{\mathbb{D}} |f_\varepsilon|^p dA$. The first derivative at 0 vanishes by symmetry, and a direct expansion (given after the proof) yields

$$J_p''(0) = \pi \left(p + \frac{p^2 \lambda^2}{p+4} + \frac{2p(p-2)\lambda}{p+2} \right). \quad (4.6)$$

The quadratic polynomial in λ on the right-hand side has minimum

$$p - \frac{(2-p)^2(p+4)}{(p+2)^2} = \frac{4(p^2 + 4p - 4)}{(p+2)^2} = \frac{4(p-p_*)(p+2+2\sqrt{2})}{(p+2)^2},$$

attained at

$$\lambda_p := \frac{(2-p)(p+4)}{p(p+2)}.$$

Since $p+2+2\sqrt{2} > 0$ for $p > 0$, the minimum is negative precisely for $p < p_*$. Fixing such a p and taking $\lambda = \lambda_p$, we have $J_p''(0) < 0$; as $J_p'(0) = 0$ and $J_p(0) = 2\pi/(p+2)$, for all sufficiently small $\varepsilon > 0$,

$$\int_{\mathbb{D}} |f_\varepsilon|^p dA < \frac{2\pi}{p+2}.$$

Thus $m_{\xi_0}(p) < 2\pi/(p+2)$, and Proposition 4.2 yields $\sup_{c \in \mathbb{C}} m_{\xi_c}(p) < 2\pi/(p+2)$. Consequently

$$\inf_{\xi \in S_H} K_{\xi, \mathbb{D}, p}(0) = \frac{1}{\sup_{c \in \mathbb{C}} m_{\xi_c}(p)} > \frac{p+2}{2\pi} = K_{\mathbb{D}}^{H, p}(0),$$

■

PROPOSITION 4.2. *The identity (4.5) holds if and only if $m_{\xi_0}(p) = 2\pi/(p+2)$, i.e. if and only if z minimizes $\int_{\mathbb{D}} |f|^p dA$ among all holomorphic f with $f'(0) = 1$.*

Proof. Since

$$\inf_{\xi \in S_H} K_{\xi, \mathbb{D}, p}(0) = \inf_{c \in \mathbb{C}} \frac{1}{m_{\xi_c}(p)} = \frac{1}{\sup_{c \in \mathbb{C}} m_{\xi_c}(p)},$$

it suffices to show that $\sup_{c \in \mathbb{C}} m_{\xi_c}(p)$ equals $2\pi/(p+2)$ if and only if it is attained at $c = 0$.

We first note that $c \mapsto m_{\xi_c}(p)$ is continuous. Write $J_p(f) := \int_{\mathbb{D}} |f|^p dA$ and

$$N(c) := \sup_{f \neq 0} \frac{|cf(0) + f'(0)|}{J_p(f)^{1/p}},$$

so that $m_{\xi_c}(p) = N(c)^{-p}$. Since $|f(0)| \leq \pi^{-1/p} J_p(f)^{1/p}$ by subharmonicity,

$$|N(c) - N(d)| \leq |c - d| \pi^{-1/p},$$

so N , and hence $c \mapsto m_{\xi_c}(p)$, is continuous. Taking $f \equiv 1$ shows $N(c) \geq |c| \pi^{-1/p}$, whence $m_{\xi_c}(p) \leq \pi |c|^{-p} \rightarrow 0$ as $|c| \rightarrow \infty$.

It remains to check that $m_{\xi_c}(p) < 2\pi/(p+2)$ for every $c \neq 0$. Set $\eta = \bar{c}/|c|$ and $f_t(z) = z + t\eta(1 - cz)$. Then $cf_t(0) + f'_t(0) = 1$ for every t , and

$$\left. \frac{d}{dt} \right|_{t=0} \int_{\mathbb{D}} |f_t|^p dA = p \Re \int_{\mathbb{D}} |z|^{p-2} \bar{z} \eta (1 - cz) dA = -p \Re(\eta c) \int_{\mathbb{D}} |z|^p dA = -p|c| \frac{2\pi}{p+2} < 0,$$

because $\int_{\mathbb{D}} |z|^{p-2} \bar{z} dA = 0$ by rotation invariance. Thus $J_p(f_t) < 2\pi/(p+2)$ for all sufficiently small $t > 0$, which gives $m_{\xi_c}(p) < 2\pi/(p+2)$.

Consequently, if $m_{\xi_0}(p) = 2\pi/(p+2)$ then the supremum is attained at $c = 0$; if instead $m_{\xi_0}(p) < 2\pi/(p+2)$, then $m_{\xi_c}(p) < 2\pi/(p+2)$ for every $c \in \mathbb{C}$, and continuity together with the decay at infinity yields $\sup_{c \in \mathbb{C}} m_{\xi_c}(p) < 2\pi/(p+2)$. $\blacksquare$

For completeness we record the second variation used above. Let $h(z) = \sum_{m \neq 1} h_m z^m$ be holomorphic with $h'(0) = 0$. Expanding $|z + \varepsilon h|^p = |z|^p |1 + \varepsilon h/z|^p$ pointwise for $z \neq 0$ and using the elementary averages $\frac{1}{2\pi} \int_0^{2\pi} e^{ik\theta} d\theta = \delta_{k,0}$, we obtain

$$\left. \frac{d^2}{d\varepsilon^2} \right|_{\varepsilon=0} \int_{\mathbb{D}} |z + \varepsilon h|^p dA = \pi \left(p^2 \sum_{m \neq 1} \frac{|h_m|^2}{p+2m} + \frac{2p(p-2)}{p+2} \Re(h_0 h_2) \right), \quad (4.7)$$

which for $h(z) = 1 + \lambda z^2$ reduces to (4.6). The only potentially negative term is the cross term $\Re(h_0 h_2)$, which arises from the pairing of the modes $h_0 z^{-1}$ and $h_2 z$ in $(h/z)^2$. The threshold p_* is exactly the value at which the two-mode quadratic form $(x, y) \mapsto px^2 + p^2 y^2/(p+4) + 2p(p-2)xy/(p+2)$ becomes positive semidefinite.

REMARK 4.3. *It remains open whether (4.5) holds for every $p \geq p_*$. For $p \geq 1$ this follows from [10, Theorem 1.7], and the numerical evidence above suggests that z remains the global minimizer for every $p \geq p_*$ in this example; a rigorous proof of (4.5) in the remaining range $p_* \leq p < 1$ is not known.*

5 Monotonicity of higher order p -Bergman kernels fails for $p > 2$

In Sections 5.1 to 5.5 we treat the family D_q of Theorem 1.2, which yields the counterexample for $p > 3$; the full range $p > 2$ is then reached in Section 5.6 by the r -fold symmetric families.

Throughout Sections 5.1 to 5.5, $p > 3$ is fixed, $q > 1$, and we use freely the notation of Section 2. By Proposition 2.1 and Proposition 2.2, $(D_c)_a = D_q$ with $q = ce^{2a}$, so the monotonicity of $a \mapsto K_{(D_c)_a}^{z^6, p}(0)$ is equivalent to that of $q \mapsto K_{D_q}^{z^6, p}(0)$.

5.1 Reduction to a weighted A^p problem on the disk

For $f \in \mathcal{O}(D_q)$ write

$$f(x) = g_e(x^2) + xg_o(x^2), \quad g_e, g_o \in \mathcal{O}(U_q),$$

with U_q as in Section 2.2. Since $q > 1$, the domain D_q is unbounded and contains a neighborhood of ∞ . Since $f \in \mathcal{O}(D_q) \cap L^p(D_q)$, near ∞ we have $f(x) = g_e(\infty) + xg_o(\infty) + o(1)$; the holomorphy of f at ∞ forces $g_o(\infty) = 0$, and then the integrability of $|f|^p$ on the unbounded domain D_q forces $g_e(\infty) = f(\infty) = 0$. Consequently, in the variable $\zeta = y/(1 - qy)$ of Lemma 2.3,

$$g_e(y(\zeta)) = (1 + q\zeta)h_e(\zeta), \quad g_o(y(\zeta)) = (1 + q\zeta)h_o(\zeta),$$

for some $h_e, h_o \in \mathcal{O}(\mathbb{D})$, because $y = \infty$ corresponds to $\zeta = -1/q$.

LEMMA 5.1. *With the notation above, the following hold.*

(i) *The jet conditions $f^{(j)}(0) = 0$ for $0 \leq j < 6$ and $f^{(6)}(0) = 1$ are equivalent to*

$$h_e(\zeta) = \frac{\zeta^3}{6!} + O(\zeta^4), \quad h_o(\zeta) = O(\zeta^3).$$

(ii) *For $b := \sqrt{\zeta/(1 + q\zeta)}$, one has*

$$\int_{D_q} |f|^p dA = \frac{1}{4} \int_{\mathbb{D}} (|h_e + bh_o|^p + |h_e - bh_o|^p) \frac{|1 + q\zeta|^{p-3}}{|\zeta|} dA(\zeta).$$

Proof. (i) follows exactly as in Lemma 2.3, since $y = \zeta + O(\zeta^2)$. For (ii), the map $x \mapsto y = x^2$ is two-to-one from D_q onto U_q with $dA(y) = 4|x|^2 dA(x) = 4|y| dA(x)$, and the two preimages of y contribute $|g_e(y) \pm wg_o(y)|^p$, $w^2 = y$. Since $w = \sqrt{\zeta/(1 + q\zeta)}$ and $dA(y) = |1 + q\zeta|^{-4} dA(\zeta)$, $|y|^{-1} = |1 + q\zeta|/|\zeta|$, we obtain

$$\int_{D_q} |f|^p dA = \frac{1}{4} \int_{\mathbb{D}} |1 + q\zeta|^p (|h_e + bh_o|^p + |h_e - bh_o|^p) |1 + q\zeta|^{-3} |\zeta|^{-1} dA(\zeta),$$

with $b := \sqrt{\zeta/(1 + q\zeta)}$, which is the claimed formula. ■

For $p \geq 1$ the function $\mathbb{C} \ni z \mapsto |z|^p$ is convex, so Jensen's inequality gives pointwise

$$\frac{1}{2} (|h_e + bh_o|^p + |h_e - bh_o|^p) \geq |h_e|^p,$$

and the odd part of f can only increase the L^p norm. Hence the infimum defining $K_{D_q}^{z^6, p}(0)$ may be taken among even functions, and by Lemma 5.1,

$$K_{D_q}^{z^6, p}(0) = 2(6!)^p M(q)^{-1},$$

where

$$M(q) := \inf \left\{ \int_{\mathbb{D}} |h|^p \frac{|1 + q\zeta|^{p-3}}{|\zeta|} dA(\zeta) : h \in \mathcal{O}(\mathbb{D}), h_3 = 1 \right\},$$

and h_3 denotes the coefficient of ζ^3 in the Taylor expansion of h at 0. For $p = 2$ this is exactly the reduction of Lemma 2.3 and Proposition 2.4: the weight $|1 + q\zeta|^{p-3}|\zeta|^{-1}$ becomes $|1 + q\zeta|^{-1}|\zeta|^{-1}$.

5.2 Lower bound

Take $h_0(\zeta) = \zeta^3/6!$, which corresponds to the even admissible function

$$f_q(x) = \frac{x^6(1 - qx^2)^{-4}}{6!},$$

because $1 + q\zeta = (1 - qy)^{-1}$ and $g_e(y) = y^3(1 - qy)^{-4}/6!$. By Lemma 5.1(ii),

$$\int_{D_q} |f_q|^p dA = \frac{1}{2(6!)^p} \int_{\mathbb{D}} |\zeta|^{3p-1} |1 + q\zeta|^{p-3} dA(\zeta) \leq \frac{\pi(1+q)^{p-3}}{(6!)^p(3p+1)},$$

since $p > 3$ gives $|1 + q\zeta|^{p-3} \leq (1+q)^{p-3}$ and $\int_{\mathbb{D}} |\zeta|^{3p-1} dA = 2\pi/(3p+1)$. Therefore

$$K_{D_q}^{z^6, p}(0) \geq \frac{(6!)^p(3p+1)}{\pi(1+q)^{p-3}} \geq c_p q^{-(p-3)}, \quad c_p := \frac{(6!)^p(3p+1)}{\pi 2^{p-3}},$$

for $q \geq 1$.

5.3 Upper bound by duality

By homogeneity,

$$M(q)^{-1} = \sup \left\{ |h_3|^p : \int_{\mathbb{D}} |h|^p w_q dA \leq 1 \right\} = \|\ell\|^p,$$

where $w_q(\zeta) := |1 + q\zeta|^{p-3}|\zeta|^{-1}$ and $\ell(h) := h_3$ on the weighted space $A^p(\mathbb{D}, w_q)$. Fix ζ_0 with $|\zeta_0| = 1/2$. Since $|h|^p$ is subharmonic for $p \geq 1$,

$$|h(\zeta_0)|^p \leq \frac{16}{\pi} \int_{B(\zeta_0, 1/4)} |h|^p dA.$$

On $B(\zeta_0, 1/4) \subset \mathbb{D}$ we have $|\zeta| \leq 3/4$, and for $q \geq 8$,

$$|1 + q\zeta| \geq q|\zeta| - 1 \geq \frac{q}{8}, \quad w_q(\zeta) \geq \frac{4}{3} \left(\frac{q}{8} \right)^{p-3}.$$

Hence

$$|h(\zeta_0)|^p \leq \frac{12}{\pi} \left(\frac{q}{8} \right)^{-(p-3)} \int_{\mathbb{D}} |h|^p w_q dA.$$

Since $h_3 = (2\pi i)^{-1} \int_{|\zeta|=1/2} h(\zeta) \zeta^{-4} d\zeta$, we have $|h_3| \leq 8 \max_{|\zeta|=1/2} |h(\zeta)|$. Thus

$$\|\ell\| \leq 8 \left(\frac{12}{\pi} \right)^{1/p} \left(\frac{q}{8} \right)^{-(p-3)/p},$$

and consequently

$$K_{D_q}^{z^6, p}(0) \leq C_p q^{-(p-3)}, \quad C_p := \frac{24(6!)^p 8^{2p-3}}{\pi}.$$

5.4 The limit as $q \rightarrow 0$

For $0 < q < 1$ we compare D_q with disks. If $|x|^2 < 1/(1+q)$, then $(1+q)|x|^2 < 1$, whence $|x|^2 < 1 - q|x|^2 \leq |1 - qx^2|$; conversely, if $x \in D_q$, then $|x|^2 < |1 - qx^2| \leq 1 + q|x|^2$, whence $|x|^2 < 1/(1-q)$. Hence

$$D\left(0, (1+q)^{-1/2}\right) \subset D_q \subset D\left(0, (1-q)^{-1/2}\right).$$

Since the kernel is non-increasing under domain inclusion and $K_{D(0,R)}^{z^6, p}(0) = R^{-(6p+2)} K_{\mathbb{D}}^{z^6, p}(0)$ by rescaling, with

$$K_{\mathbb{D}}^{z^6, p}(0) = \frac{(6!)^p (6p+2)}{2\pi},$$

we get

$$(1-q)^{3p+1} \frac{(6!)^p (6p+2)}{2\pi} \leq K_{D_q}^{z^6, p}(0) \leq (1+q)^{3p+1} \frac{(6!)^p (6p+2)}{2\pi},$$

and $K_{D_q}^{z^6, p}(0) \rightarrow (6!)^p (6p+2)/(2\pi)$ as $q \rightarrow 0^+$.

5.5 Proof of Theorem 1.5

By Sections 5.2 and 5.3,

$$c_p q^{-(p-3)} \leq K_{D_q}^{z^6, p}(0) \leq C_p q^{-(p-3)}, \quad q \geq 8,$$

so $K_{D_q}^{z^6, p}(0) \rightarrow 0$ as $q \rightarrow \infty$, while Section 5.4 gives $K_{D_q}^{z^6, p}(0) \rightarrow (6!)^p (6p+2)/(2\pi) > 0$ as $q \rightarrow 0^+$. Choose $c > 1$ so large that $C_p c^{-(p-3)} < (1 - c^{-1})^{3p+1} (6!)^p (6p+2)/(2\pi)$. Then

$$K_{D_c}^{z^6, p}(0) \leq C_p c^{-(p-3)} < (1 - c^{-1})^{3p+1} \frac{(6!)^p (6p+2)}{2\pi} \leq K_{D_{1/c}}^{z^6, p}(0).$$

By Proposition 2.2, $D_{1/c} = D_{ce^{2a}}$ with $a = -\log c$, i.e. $D_{1/c} = (D_c)_{-\log c}$, while $D_c = (D_c)_0$. Thus $-\log c < 0$ but $K_{(D_c)_{-\log c}}^{z^6, p}(0) > K_{(D_c)_0}^{z^6, p}(0)$, and $a \mapsto K_{(D_c)_a}^{z^6, p}(0)$ is not non-decreasing. This proves the theorem.

Finally, the Green function formula of Proposition 2.1 gives $G_{D_c}(z, 0) = \log |z| + O(1)$ near 0, so the Azukawa indicatrix of D_c at 0 is the unit disk $\mathbb{D}$, and the limit

$$\lim_{q \rightarrow 0^+} K_{D_q}^{z^6, p}(0) = \frac{(6!)^p (6p+2)}{2\pi} = K_{\mathbb{D}}^{z^6, p}(0)$$

holds with equality.

5.6 Extension to every $p > 2$ by r -fold symmetry

Fix integers $r \geq 2$ and $m \geq 1$, set $H(z) = z^{rm}$, and consider

$$D_q^{(r)} := \{x \in \mathbb{C} : |x|^r < |1 - qx^r|\}, \quad q > 1.$$

The map

$$F_q(x) := \frac{x^r}{1 - qx^r}$$

maps $D_q^{(r)}$ properly, r -to-one, onto $\mathbb{D} \setminus \{-1/q\}$ and vanishes to order r at 0, so the Green function of $D_q^{(r)}$ with pole at 0 is

$$G_{D_q^{(r)}}(x, 0) = \frac{1}{r} \log |F_q(x)| = \log |x| - \frac{1}{r} \log |1 - qx^r|,$$

and, exactly as in Proposition 2.2,

$$(D_c^{(r)})_a = D_q^{(r)}, \quad q = ce^{ra}.$$

Let $f \in \mathcal{O}(D_q^{(r)}) \cap L^p(D_q^{(r)})$ satisfy $f^{(j)}(0) = 0$ for $0 \leq j < rm$ and $f^{(rm)}(0) = 1$, and let $\omega := e^{2\pi i/r}$. Averaging f over the r -th roots of unity,

$$f_0(x) := \frac{1}{r} \sum_{j=0}^{r-1} f(\omega^j x),$$

we get $f_0(\omega x) = f_0(x)$, hence $f_0(x) = g(x^r)$ with $g \in \mathcal{O}(U_q)$. Convexity gives $\|f_0\|_{L^p} \leq \|f\|_{L^p}$, and since $\omega^{jrm} = 1$, the average f_0 satisfies the same jet conditions as f . Thus the extremal problem for $K_{D_q^{(r)}}^{z^{rm}, p}(0)$ may be restricted to r -fold symmetric functions. As in Theorem 5.1, the integrability forces $g(\infty) = 0$, whence $g(y(\zeta)) = (1 + q\zeta)h(\zeta)$ with $h \in \mathcal{O}(\mathbb{D})$ and $h(\zeta) = \zeta^m/(rm)! + O(\zeta^{m+1})$.

Since $dA(y) = r^2|x|^{2r-2}dA(x) = r^2|y|^{2-2/r}dA(x)$ and every y has exactly r preimages, each contributing $|g(y)|^p$, we obtain

$$\int_{D_q^{(r)}} |f|^p dA = \frac{1}{r} \int_{U_q} |g(y)|^p |y|^{2/r-2} dA(y) = \frac{1}{r} \int_{\mathbb{D}} |h(\zeta)|^p |\zeta|^{2/r-2} |1 + q\zeta|^{p-2-2/r} dA(\zeta),$$

using $dA(y) = |1 + q\zeta|^{-4}dA(\zeta)$ and $|y|^{2/r-2} = |\zeta|^{2/r-2}|1 + q\zeta|^{2-2/r}$. Therefore

$$K_{D_q^{(r)}}^{z^{rm}, p}(0) = r(rm)!^p M(q)^{-1}, \quad M(q) := \inf \left\{ \int_{\mathbb{D}} |h|^p w_q dA : h \in \mathcal{O}(\mathbb{D}), h_m = 1 \right\},$$

where $w_q(\zeta) := |\zeta|^{2/r-2}|1 + q\zeta|^{p-2-2/r}$ and h_m is the coefficient of ζ^m . This is the natural r -fold analogue of the reduction of Section 5.1; for $r = 2$, $m = 3$ it recovers Theorem 5.1 with $H(z) = z^6$.

Lower bound. Assume $p > 2 + 2/r$, so that $\beta := p - 2 - 2/r > 0$. Take $h_0(\zeta) = \zeta^m$, which corresponds to the symmetric admissible function

$$f_q(x) = \frac{x^{rm}(1 - qx^r)^{-(m+1)}}{(rm)!},$$

Since $|1 + q\zeta| \leq 1 + q$ on $\mathbb{D}$ and $\int_{\mathbb{D}} |\zeta|^{pm+2/r-2} dA = 2\pi/(pm + 2/r)$,

$$M(q) \leq \int_{\mathbb{D}} |\zeta|^{pm+2/r-2} |1 + q\zeta|^\beta dA \leq \frac{2\pi(1+q)^\beta}{pm + 2/r},$$

hence

$$K_{D_q^{(r)}}^{z^{rm},p}(0) \geq \frac{r(rm)!^p(pm + 2/r)}{2\pi(1+q)^\beta} \geq c_{p,r,m} q^{-\beta}$$

for $q \geq 1$.

Upper bound. Let $\ell(h) := h_m$ on $A^p(\mathbb{D}, w_q)$. By homogeneity $M(q)^{-1} = \|\ell\|^p$. Fix $|\zeta_0| = 1/2$. Subharmonicity gives $|h(\zeta_0)|^p \leq \frac{16}{\pi} \int_{B(\zeta_0, 1/4)} |h|^p dA$. On $B(\zeta_0, 1/4)$ we have $|\zeta| \leq 3/4$; since $2/r - 2 \leq 0$, $|\zeta|^{2/r-2} \geq (3/4)^{2/r-2}$, and for $q \geq 8$, $|1 + q\zeta| \geq q/8$. Hence

$$w_q(\zeta) \geq \left(\frac{3}{4}\right)^{2/r-2} \left(\frac{q}{8}\right)^\beta,$$

and

$$|h(\zeta_0)|^p \leq \frac{16}{\pi} \left(\frac{3}{4}\right)^{2-2/r} \left(\frac{q}{8}\right)^{-\beta} \int_{\mathbb{D}} |h|^p w_q dA.$$

Since $h_m = (2\pi i)^{-1} \int_{|\zeta|=1/2} h(\zeta) \zeta^{-m-1} d\zeta$, one has $|h_m| \leq 2^m \max_{|\zeta|=1/2} |h(\zeta)|$, and consequently

$$K_{D_q^{(r)}}^{z^{rm},p}(0) \leq C_{p,r,m} q^{-\beta}, \quad q \geq 8.$$

The limit as $q \rightarrow 0$. For $0 < q < 1$, exactly as in Section 5.4,

$$D\left(0, (1+q)^{-1/r}\right) \subset D_q^{(r)} \subset D\left(0, (1-q)^{-1/r}\right),$$

and rescaling gives $K_{D(0,R)}^{z^{rm},p}(0) = R^{-(2+prm)} K_{\mathbb{D}}^{z^{rm},p}(0)$ with $K_{\mathbb{D}}^{z^{rm},p}(0) = (rm)!^p(rmp + 2)/(2\pi)$. Therefore

$$\lim_{q \rightarrow 0^+} K_{D_q^{(r)}}^{z^{rm},p}(0) = \frac{(rm)!^p(rmp + 2)}{2\pi},$$

and the Green function formula above gives $I_{D_c^{(r)}}(0) = \mathbb{D}$, so the comparison with the Azukawa indicatrix holds with equality.

Proof of Theorem 1.6. For $p > 2$ choose an integer $r > 2/(p-2)$, so that $\beta = p - 2 - 2/r > 0$, and take any $m \geq 1$. The two bounds above give

$$c_{p,r,m} q^{-\beta} \leq K_{D_q^{(r)}}^{z^{rm},p}(0) \leq C_{p,r,m} q^{-\beta}, \quad q \geq 8,$$

so the kernel tends to 0 as $q \rightarrow \infty$, while its limit at $q \rightarrow 0^+$ is the positive constant $(rm)!^p(rmp + 2)/(2\pi)$. Choose $c > 1$ so large that $C_{p,r,m} c^{-\beta} < (1 - c^{-1})^{(2+prm)/r} (rm)!^p(rmp + 2)/(2\pi)$; then

$$K_{D_c^{(r)}}^{z^{rm},p}(0) \leq C_{p,r,m} c^{-\beta} < (1 - c^{-1})^{(2+prm)/r} \frac{(rm)!^p(rmp + 2)}{2\pi} \leq K_{D_{1/c}^{(r)}}^{z^{rm},p}(0).$$

Since $q = ce^{ra}$, the identity $q = 1/c$ corresponds to $a = -\frac{1}{r} \log c$, i.e. $D_{1/c}^{(r)} = (D_c^{(r)})_{-\frac{1}{r} \log c}$, while $D_c^{(r)} = (D_c^{(r)})_0$. Thus the kernel at the smaller value $a = -\frac{1}{r} \log c < 0$ exceeds its value at $a = 0$, and $a \mapsto K_{(D_c^{(r)})_a}^{z^{rm},p}(0)$ is not non-decreasing. $\blacksquare$

6 The L^p concavity of minimal integrals fails for $p > 2$

Throughout this section we fix $p > 2$ and integers $r \geq 1$, $m \geq 1$, and set $k := rm$. We fix $c \in (0, 1)$ and write

$$D := D_c^{(r)} = \{x \in \mathbb{C} : |x|^r < |1 - cx^r|\}, \quad F_c(x) := \frac{x^r}{1 - cx^r}.$$

Since $c < 1$, the domain D is bounded: for $x \in D$,

$$|x|^r < |1 - cx^r| \leq 1 + c|x|^r,$$

so $|x| < (1 - c)^{-1/r}$.

6.1 The multiplier ideal of the Green function

Exactly as in Theorem 2.1 (whose proof applies verbatim to the map F_c), the map $F_c : D \rightarrow \mathbb{D} \setminus \{-1/c\}$ is proper of degree r , it vanishes to order r at 0, and consequently

$$G_D(x, 0) = \frac{1}{r} \log |F_c(x)| = \log |x| - \frac{1}{r} \log |1 - cx^r|$$

is the Green function of D with pole at 0. Set

$$\varphi := 2(k+1)G_D(\cdot, 0), \quad F(x) := \frac{x^k}{k!}.$$

Then φ is a negative plurisubharmonic function on D with $\varphi(0) = -\infty$, and near 0,

$$e^{-\varphi} = |x|^{-2(k+1)} |1 - cx^r|^{2(k+1)/r} = |x|^{-2(k+1)} (1 + O(|x|^r)).$$

LEMMA 6.1. *One has $\mathcal{I}(\varphi)_0 = \mathfrak{m}_0^{k+1}$.*

Proof. For $f \in \mathcal{O}_0$ with $\text{ord}_0 f = j$, one has $|f|^2 e^{-\varphi} \asymp |x|^{2j-2(k+1)}$ near 0, and in polar coordinates

$$\int_{|x| < \delta} |f|^2 e^{-\varphi} dA \asymp \int_0^\delta s^{2j-2k-1} ds,$$

which is finite exactly when $j \geq k+1$. ■

6.2 Reduction to a higher order p -Bergman kernel

Let $\Omega \subset \mathbb{C}$ be a domain containing 0. Since $(\tilde{F} - F, 0) \in \mathfrak{m}_0^{k+1}$ is equivalent to

$$\tilde{F}^{(j)}(0) = 0 \quad (0 \leq j < k), \quad \tilde{F}^{(k)}(0) = 1,$$

Lemma 6.1 gives

$$C_{F, \mathcal{I}(\varphi)_0, p}(\Omega) = \inf \left\{ \int_\Omega |\tilde{F}|^p : \tilde{F} \in \mathcal{O}(\Omega), \tilde{F}^{(j)}(0) = 0 \quad (0 \leq j < k), \tilde{F}^{(k)}(0) = 1 \right\} = \frac{1}{K_\Omega^{z^k, p}(0)},$$

where $K_\Omega^{z^k, p}(0)$ is the higher order p -Bergman kernel of Section 4. Indeed, every admissible $\tilde{F}$ satisfies $\|\tilde{F}\|_{L^p(\Omega)}^p \geq 1/K_\Omega^{z^k, p}(0)$ by the definition of the kernel, and rescaling functions with $f^{(j)}(0) = 0$ ($0 \leq j < k$) and $f^{(k)}(0) \neq 0$ by $1/f^{(k)}(0)$ produces admissible functions with norms arbitrarily close to $1/K_\Omega^{z^k, p}(0)$.

6.3 Sublevel sets and the scaling law

For $t \geq 0$ set

$$a := -\frac{t}{2(k+1)}, \quad q := ce^{ra}.$$

As in Theorem 2.2 (applied to the r -fold map F_c , exactly as in Section 5.6),

$$\{G_D(\cdot, 0) < a\} = e^a D_q^{(r)}, \quad D_q^{(r)} := \{x \in \mathbb{C} : |x|^r < |1 - qx^r|\} :$$

indeed $G_D(z, 0) < a$ is equivalent to $|F_c(z)| < e^{ra}$, and writing $z = e^a x$ gives $F_c(e^a x) = e^{ra} F_q(x)$.

LEMMA 6.2. *For every $t \geq 0$,*

$$G_p(t) = \frac{e^{(pk+2)a}}{K_{D_q^{(r)}}^{z^k, p}(0)}, \quad a = -\frac{t}{2(k+1)}, \quad q = ce^{ra}.$$

Proof. Since $G_D(\cdot, 0) < 0$ on D ,

$$\{\varphi < -t\} \cap D = \{G_D(\cdot, 0) < a\} = e^a D_q^{(r)}.$$

For $\tilde{F} \in \mathcal{O}(e^a D_q^{(r)})$ set $g(x) := \tilde{F}(e^a x) \in \mathcal{O}(D_q^{(r)})$. Then $g^{(j)}(0) = e^{aj} \tilde{F}^{(j)}(0)$, so the admissibility of $\tilde{F}$ is equivalent to $g^{(j)}(0) = 0$ ($0 \leq j < k$) and $g^{(k)}(0) = e^{ak}$, while

$$\int_{e^a D_q^{(r)}} |\tilde{F}|^p dA = e^{2a} \int_{D_q^{(r)}} |g|^p dA.$$

By the homogeneity of the previous subsection,

$$G_p(t) = e^{2a} \cdot \frac{e^{pak}}{K_{D_q^{(r)}}^{z^k, p}(0)} = \frac{e^{(pk+2)a}}{K_{D_q^{(r)}}^{z^k, p}(0)}.$$

■

6.4 The limit as $t \rightarrow +\infty$ and the failure of concavity

As computed in Section 5.6 — the comparison

$$D(0, (1+q)^{-1/r}) \subset D_q^{(r)} \subset D(0, (1-q)^{-1/r}), \quad 0 < q < 1,$$

together with the rescaling $K_{D(0,R)}^{z^k, p}(0) = R^{-(pk+2)} K_{\mathbb{D}}^{z^k, p}(0)$ and the disk value $K_{\mathbb{D}}^{z^k, p}(0) = (k!)^p (kp+2)/(2\pi)$ — one has, for every $p \geq 1$,

$$\lim_{q \rightarrow 0^+} K_{D_q^{(r)}}^{z^k, p}(0) = \frac{(k!)^p (kp+2)}{2\pi} =: L.$$

Proof of Theorem 1.7. First, $G_p(0) < +\infty$: since $(F - F, 0) \in \mathfrak{m}_0^{k+1}$ and $F \in L^p(D)$ (D is bounded), we have $G_p(0) \leq \|F\|_{L^p(D)}^p < +\infty$. Moreover $G_p(0) > 0$: every admissible $\tilde{F}$ has

$\tilde{F}^{(k)}(0) = 1$, hence positive L^p norm on a neighborhood of 0; equivalently $G_p(0) = 1/K_D^{z^k, p}(0) \in (0, +\infty)$.

As $t \rightarrow +\infty$ we have $a \rightarrow -\infty$ and $q = ce^{ra} \rightarrow 0$, so Lemma 6.2 and the limit above give

$$G_p(t) \sim \frac{1}{L} e^{(pk+2)a} = \frac{1}{L} \exp\left(-\frac{pk+2}{2(k+1)} t\right).$$

Let $g(\rho) := G_p(-\log \rho)$ for $\rho \in (0, 1]$. Then

$$g(\rho) \sim \frac{1}{L} \rho^\mu, \quad \mu := \frac{pk+2}{2(k+1)} = 1 + \frac{k(p-2)}{2(k+1)} > 1,$$

as $\rho \rightarrow 0^+$.

The function g is non-decreasing, because G_p is non-increasing in t . Moreover $\lim_{t \rightarrow +\infty} G_p(t) = 0$, hence $g(0+) = 0$: indeed,

$$G_p(t) \leq \int_{\{\varphi < -t\}} |F|^p dA \rightarrow 0$$

by dominated convergence as the sublevel sets shrink to $\{0\}$. Also $g(1) = G_p(0) > 0$.

If g were concave on $(0, 1]$, then for $0 < \varepsilon < \rho < 1$,

$$g(\rho) \geq \frac{\rho - \varepsilon}{1 - \varepsilon} g(1) + \frac{1 - \rho}{1 - \varepsilon} g(\varepsilon),$$

and letting $\varepsilon \rightarrow 0^+$ gives $g(\rho) \geq \rho g(1)$ for every $\rho \in (0, 1]$. But $\mu > 1$ and $g(\rho) \sim \rho^\mu/L$ imply $g(\rho) < \rho g(1)$ for all sufficiently small ρ — a contradiction. Hence $g = G_p(-\log \rho)$ is not concave in $\rho \in (0, 1]$, which proves the theorem. $\blacksquare$

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